

EdgeCDSS v3.0 — Goals and Technology

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Overview

EdgeCDSS v3.0 is the third iteration of the hybrid cloud-edge clinical decision support system developed under the AI in Austere Medicine Project. Version 3.0 is defined by three primary objectives: establishing a formal clinical safety architecture, introducing a full voice interface for field use, and expanding patient context awareness to support longitudinal care management within a single patient encounter.

v3.0 is driven directly by findings from the v2.5 evaluation period, which identified three P1 patient safety failures through real-world external testing. These failures — a pediatric ketamine overdose, a missing surgical airway recommendation in a cannot-intubate scenario, and inappropriate high-flow oxygen in a COPD patient — were not detected by the automated test suite and were only discovered through clinical expert testing. v3.0 addresses these findings at the architectural level, not just through prompt adjustments.

Version Goals

Goal 1 — Clinical Safety Architecture

Implement a two-pass response generation pipeline in which every LLM-generated response passes through a secondary validation layer before reaching the provider. The validator is a separate LLM call with a focused safety-checking system prompt. It checks for pediatric dosing ceiling violations, contraindicated drug recommendations, route-context mismatches, and CICO pathway failures. If the validator flags a safety issue it either blocks the response and requests a revision or appends a clinical warning.

This architecture mirrors the safety validation pattern used in commercial-grade clinical AI systems and addresses the root cause of the v2.5 pediatric overdose finding — not prompt wording, but the absence of an independent verification step.

Goal 2 — Patient Context and Memory System

Expand the rolling 5-exchange conversation memory from v2.5 into a structured patient record. At session start, the provider optionally completes a brief patient intake — weight, age, mechanism, estimated time of injury, and available medications. This record is injected into every subsequent query, ensuring the system always has the full clinical picture. The medication inventory feature addresses the v2.5 finding where providers gave inventory context that the system ignored.

Scope of practice detection is added in this version. When a provider states their certification level (BLS, EMT, Paramedic, Critical Care) the system adjusts recommendation scope accordingly, restricting interventions to those within the stated training level.

Goal 3 — Voice Interface

Introduce a production-quality voice interface for field use. v3.0 ships two interfaces sharing the same backend: a text-only web interface for continued evaluation and feedback collection, and a voice-enabled interface using ElevenLabs TTS for demo and serious field testing. Voice commands — stop, repeat, slow down, new patient — are implemented on both the web voice interface and the Radxa edge client. Wake word detection (HEY MEDIC) is added to the Radxa client.

Goal 4 — Clinical Protocol Expansion

Address the clinical gaps identified in v2.5 evaluation. Pulmonary embolism is added as a dedicated protocol — it was entirely absent from the v2.5 knowledge base. CICO management, BLS-appropriate care pathways, route selection from scene context, and household chemical ingestion protocols are all added. The sepsis/DCR differentiation rule from v2.4 is strengthened with harder criteria after continued failures in v2.5 testing.

Goal 5 — Evaluation Infrastructure

Expand the automated test suite from 34 to 50+ cases, adding pediatric RSI, CICO, COPD oxygen management, pulmonary embolism, BLS scope, and route selection scenarios. Introduce structured tester intake to stratify feedback by clinical background. Make feedback submission required after each response. Add post-session survey after 5+ queries. Target pass rate: 92%+.

Success Criteria

Criterion	Target	Method
Automated test pass rate	92%+	Expanded 50+ case test suite
Dosing errors (any patient)	0	Safety validator hard caps — adult and pediatric
CICO surgical airway	100% mention rate	Dedicated CICO rule
COPD oxygen management	100% correct	SpO2 target rule
Sepsis/DCR differentiation	100% correct	Hard etiology rule
Mean response time	<4 seconds	GPT-4o-mini + validator
Validator overhead	<1.5 seconds	Parallel or fast LLM call
Voice interface latency	<5 seconds total	Streaming TTS

Technology Stack

Cloud Infrastructure

Component	Technology	Purpose
Cloud VM	GCP e2-medium (arcaneone)	Backend host — FastAPI + ChromaDB
Reverse proxy	nginx + Let's Encrypt SSL	HTTPS termination, maintenance fallback
DNS	DuckDNS dynamic DNS	arcaneone.duckdns.org public endpoint
Process management	systemd (cdss.service)	Auto-restart, boot persistence
Deployment	wget from GitHub raw URLs	One-command server update

Backend Application

Component	Technology	Purpose
API framework	FastAPI (Python)	REST endpoints, async request handling
Vector database	ChromaDB	Semantic search over JTS CPG knowledge base
Embeddings	OpenAI text-embedding-ada-002	7,186 indexed protocol chunks
Primary LLM	GPT-4o-mini	Clinical response generation, 2.8s avg
Safety validator	GPT-4o-mini (second call)	NEW v3.0 — independent safety check pass
TTS	ElevenLabs API (server-side)	Audio generation, MP3 streaming response
Authentication	X-Access-Token header	Shared secret on all endpoints
Feedback logging	Flat file (feedback.log)	Append-only server-side log
Dependency management	pip + venv	Isolated Python environment
HTTP client	httpx (async)	ElevenLabs and external API calls

New v3.0 Backend Components

Component	Technology	Purpose
Safety validator	GPT-4o-mini second pass	Checks dosing, contraindications, route, CICO
Patient record store	In-memory session dict	Weight, age, meds, vitals, mechanism per session
Medication inventory	Session-scoped JSON	Provider-loaded available meds list
Scope detector	Prompt-level classification	BLS/EMT/Paramedic/CC scope adjustment
Structured intake	Web form + voice prompt	Patient record population at session start

Edge Hardware

Device	Role	Specs
Radxa Zero 3W	Primary field terminal	ARM64, ~2W, voice I/O, TTS, thin client
Raspberry Pi 4	Development and backup	Full audio subsystem, testing platform
Jetson Orin Nano	Planned — offline LLM host	Next project — local inference, no internet

Voice and Audio

Component	Technology	Purpose
TTS engine	ElevenLabs API (voice interface)	High-quality field voice output
TTS fallback	Web Speech API (text interface)	Browser-native, no token cost
STT	Web Speech API (browser)	Voice-to-text query input
STT (edge)	Vosk or OpenWakeWord	Offline wake word + STT on Radxa

Audio output	pygame (Radxa)	MP3 playback on edge device
Voice commands	Keyword detection	Stop, repeat, slow down, new patient
Wake word	OpenWakeWord	HEY MEDIC trigger on Radxa client

Web Interface and DevOps

Component	Technology	Purpose
Web interface	Vanilla HTML/CSS/JS	No framework — GitHub Pages hosted
Hosting	GitHub Pages	Static hosting, auto-deploy on push
Version control	Git + GitHub	AI-in-Austere-Medicine-Project org
CI/CD	GitHub Pages auto-deploy	Push to main = live in minutes
Monitoring	journalctl + feedback.log	Query volume and clinical gap tracking
Test suite	Python requests + pytest	50+ case automated clinical evaluation
Documentation	ReportLab PDF + Markdown	Evaluation reports, PoC papers, roadmaps

v3.0 Architecture — Two-Pass Safety Pipeline

The core architectural change in v3.0 is the introduction of a secondary validation pass. Every query now travels through two LLM calls before the response reaches the provider:

Step	Component	Action
1	Provider	Voice or text query submitted with patient record and medication inventory
2	ChromaDB	Semantic search retrieves relevant JTS protocol chunks
3	GPT-4o-mini (Pass 1)	Generates clinical response using system prompt + RAG context + patient record
4	Safety Validator (Pass 2)	NEW — Independent LLM checks dosing, contraindications, CICO, route, scope
5	Gate	If validator flags issue: revise or append warning. If clean: pass through.
6	Provider	Validated response delivered via text or ElevenLabs TTS

The validator checks for: pediatric dosing ceiling violations (ketamine >2mg/kg, rocuronium >1.2mg/kg), contraindicated drugs (adenosine in WPW, high-flow O2 in COPD), CICO pathway (failed ETT + failed supraglottic = cric required), route mismatch (combative patient receiving IV dose), and DCR triggered for septic presentation. Estimated validator overhead: 0.5-1.5 seconds.

Build Timeline

Phase	Focus	Exit Criteria
Phase 1	Safety architecture — validator, P1 fixes, test suite expansion	All P1 safety cases pass validator. 90%+ on expanded suite.
Phase 2	Patient context — structured record, inventory, scope detection	Inventory queries answered correctly. BLS scope respected.

Phase 3	Voice interface — ElevenLabs, wake word, voice commands	Full voice loop tested end-to-end on Radxa.
Phase 4	Clinical gaps — PE, CICO, BLS pathway, route selection	All new protocol test cases passing.
Phase 5	Evaluation and close-out — 2-week public test, final report	92%+ pass rate. v3.0 tagged and closed on GitHub.

Research prototype — not validated for clinical use — not for patient care decisions | github.com/Al-in-Austere-Medicine-Project/pi-cloud-cdss | Open source